

LAB EXPERIMENT RECORD

01-10-2026--Set 6: Product Sales Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	01 OCTOBER 2026
Day	Thursday

Aim

To analyze the first-quarter sales of each product using a grouped bar chart, stacked area chart and table, and to build an interactive Tableau dashboard for exploring the sales data.

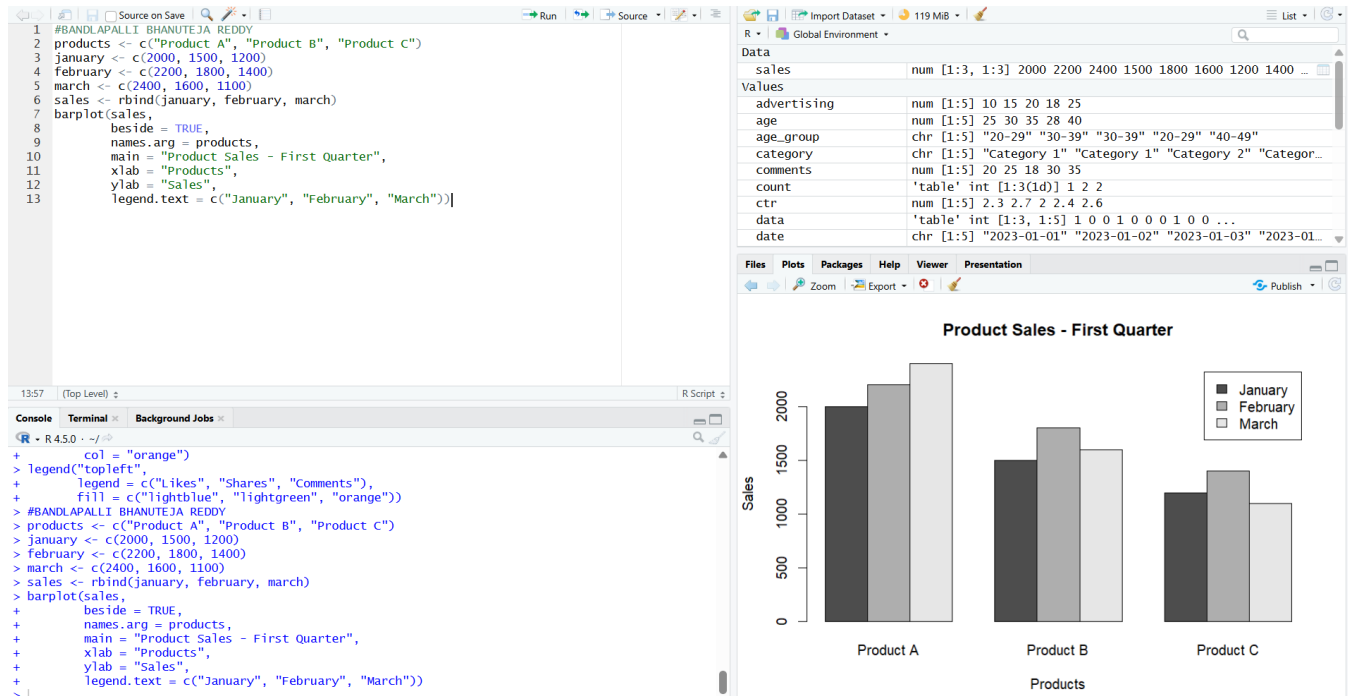
Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Question 1: Grouped Bar Chart of First-Quarter Sales

Question: Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

Aim: To compare the January, February and March sales of each product using a labelled grouped bar chart.

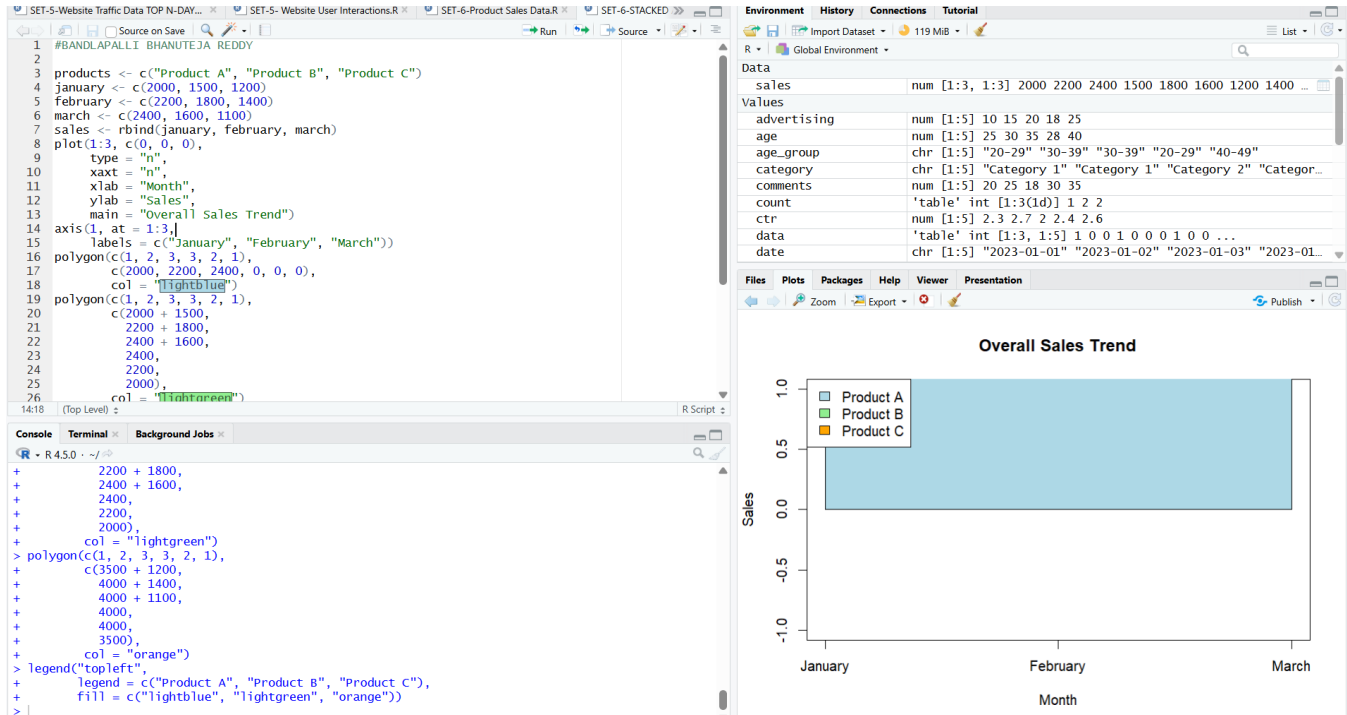


Result: Product A has the highest sales in every month, rising from 2000 to 2400, while Products B and C peak in February.

Question 2: Stacked Area Chart of Overall Sales Trend

Question: Generate a stacked area chart to represent the overall sales trend for all products over the three months.

Aim: To represent the overall sales trend of all products over the three months using a stacked area chart.



Result: Combined sales rise from 4700 in January to a peak of 5400 in February, then fall to 5100 in March.

Question 3: Table of Monthly Product Sales

Question: Build a table to show the monthly sales data for each product. Label the table elements.

Aim: To present the monthly sales of each product in a labelled table.

The screenshot shows the R Studio interface with four tabs: 'SET-5- Website User Interactions.R', 'SET-6-Product Sales Data.R', 'SET-6-STACKED AREA CHART.R', and 'SET-6-SALES TABLE.R'. The 'SET-6-SALES TABLE.R' tab is active, displaying the following R code:

```

1 #BANDLAPALLI BHANUTEJA REDDY
2 product_id <- c(1, 2, 3)
3 product_name <- c("Product A",
4                   "Product B",
5                   "Product C")
6 january <- c(2000, 1500, 1200)
7 february <- c(2200, 1800, 1400)
8 march <- c(2400, 1600, 1100)
9 sales_table <- data.frame(
10   Product_ID = product_id,
11   Product_Name = product_name,
12   January_Sales = january,
13   February_Sales = february,
14   March_Sales = march
15 )
16 print(sales_table)

```

The console output shows the execution of the code, resulting in a data frame table:

```

+           "Product B",
+           "Product C")
> january <- c(2000, 1500, 1200)
> february <- c(2200, 1800, 1400)
> march <- c(2400, 1600, 1100)
> sales_table <- data.frame(
+   Product_ID = product_id,
+   Product_Name = product_name,
+   January_Sales = january,
+   February_Sales = february,
+   March_Sales = march
+ )
> print(sales_table)
  Product_ID Product_Name January_Sales February_Sales March_Sales
1          1   Product A         2000          2200          2400
2          2   Product B         1500          1800          1600
3          3   Product C         1200          1400          1100
> |

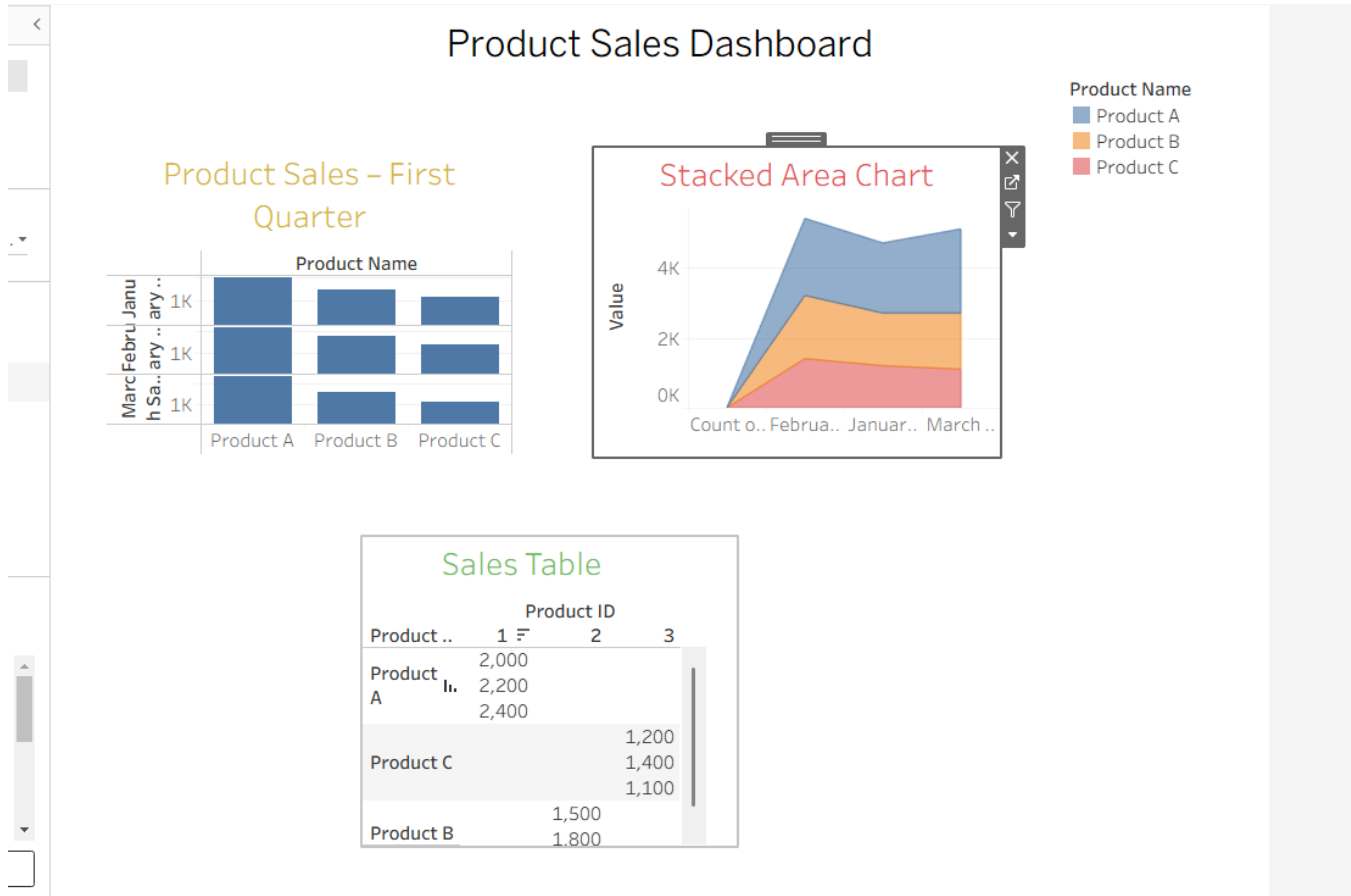
```

Result: The table lists each product's monthly sales; Product A totals 6600, Product B 4900 and Product C 3700 for the quarter.

Question 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Aim: To build an interactive Tableau dashboard that combines the grouped bar chart, stacked area chart and table for exploring sales data.



Result: The dashboard places all three views together, and filters on product and month let users explore sales interactively.

Overall Result

Observation: Total first-quarter sales were 15200, of which Product A contributed the most (6600), and February was the best month overall (5400).